



Society of Petroleum Engineers

SPE-229194-MS

Development of the H₂S Field Navigator Smartphone Application

G. Basile, M. Al Asimi, H. Al Jufaili, S. Al Rawahi, S. Al Maskari, W. Al Ramadhani, and A. Al Mahrouqi, Occidental Oil & Gas Corporation, Muscat, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229194-MS](https://doi.org/10.2118/229194-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The project ensures field operators have seamless access to the most up-to-date data regarding the concentration of Hydrogen Sulfide (H₂S) within well heads, as well as the potential extension of risk areas or H₂S zones on well pads and process facilities, through their smartphones. The use of risk areas or H₂S zones is a well-established practice in the oil and gas industry. However, the resulting maps are typically confined to risk study reports or, at best, to field posters. The "H₂S Field Navigator" application revolutionizes this by putting relevant and current H₂S data directly in operators' hands.

Periodic H₂S field measurement surveys data were consolidated into a centralized database, supported by interactive data visualization software. An internally developed application, utilizing a low-code development tool, identified well head locations on a web mapping platform based on their coordinates. This application accessed both the centralized measurement database and a separate database containing H₂S dispersion modeling results. The Radius of Exposure (ROE) was determined through empirical dispersion modeling using the highest historical readings. The application updates every four hours to keep data current.

The associated gas in an onshore heavy oil field operated by Oxy Oman, encompassing over 3,000 development wells across multiple reservoirs, exhibits a wide range of H₂S concentrations at wellhead level. The "H₂S Field Navigator" smartphone application enables users to select a specific well and provides the H₂S concentration range based on the highest historical H₂S readings for that well. In the absence of H₂S readings, the application provides a cautionary warning to operators. Additionally, the application visually maps risk areas or H₂S zones based on ROE and H₂S classification of process facilities. According to Oxy Oman's policies, each zone necessitates distinct safety controls. This is particularly important for the field, which is characterized by wells and reservoirs with a wide variety of risk profiles. The utilization of this smartphone application enhances the implementation of risk-based H₂S policies, improves field operators' access to critical safety information, and ultimately fosters a H₂S culture grounded in awareness and preparedness.

Introduction

The primary objective of this project was to ensure that field operators have seamless access to up-to-date data regarding the concentration of H₂S within wellheads, as well as the potential extension of risk areas or H₂S zones on both well pads and process facilities, through their smartphones. The use of risk areas or H₂S zones is a well-established practice in the oil and gas industry. However, the resulting maps are typically confined to risk study reports or, at best, to field posters. The "H₂S Field Navigator" application revolutionizes this approach by providing the most up-to-date and relevant information directly on smartphones, ensuring immediate accessibility for field operators.

Field Context and Challenges

Oxy Oman operates an extensive onshore heavy oil field in Oman, which encompasses over 3,000 development wells distributed across multiple reservoirs. To facilitate the extraction of oil, steam is injected into the reservoir. Initially, the main reservoir was classified as sweet; however, over time, elevated amounts of H₂S were generated within the reservoir through a process known as Aqua-thermolysis. As a result, the wells now exhibit a wide range of H₂S concentrations, varying from 0 to 50,000 ppm in the gas phase.

System Architecture and Data Integration

The development of the H₂S Field Navigator involved the integration of various components and datasets, which are crucial for its functionality. These components include:

1. The Navigation Tool.
2. The H₂S Measurements database.
3. H₂S ROE calculations.

The following sections provide more detailed information on these vital constituents and their integration into the H₂S Field Navigator application.

The development of the Navigation Tool started in November 2023, while the integration of the H₂S Measurements database and the H₂S ROE calculations was established in November 2024. In the following months tests were conducted by selected users, before the release to the entire Company personnel during the first week of August 2025.

The Navigation tool. The Field Navigator component of the application has been developed as an integrated digital platform. This innovative tool empowers field teams to swiftly identify and access all field locations, including wells, stations, and key equipment, by leveraging GPS coordinates and integrating them with a digital mapping platform. The application was internally developed using flexible tools, granting full control over its design and scalability to meet evolving needs. Featuring a user-friendly and practical interface, the application enables users to efficiently identify and reach target locations, thereby contributing to reduced time expenditure and improved daily operations.

The application addresses the inefficiencies and heightened risks associated with manually searching for locations in the field.

The application boasts a simple and intuitive main interface. Users can select the type of location they want to find, such as wells, process facilities, or assembly points. A search bar is provided for entering the name of the well or site. Upon selecting the desired result, the location is instantly displayed on the integrated map. The app offers route navigation options, including the fastest and shortest paths through a digital mapping platform. Additionally, it provides a 3-dimensional map preview for assessing road conditions and identifying potential shortcuts. Voice-guided directions further enhance navigation safety and convenience.

Establishment of an H₂S Measurements Database. The creation of a comprehensive H₂S Measurements database was a pivotal component for the successful execution of the project, providing seamless access to both recent and historical data. Data was meticulously sourced from two primary channels: well measurements, encompassing casing and wellhead data, and facility-based recordings.

The team undertook the systematic sampling of approximately 35 wells daily, diligently documenting critical information such as sampling location, H₂S concentration (ppm), and the sampling date. This rigorous activity supported the broader objective of sampling every designated well at least once per quarter. Given the extensive volume of well sampling data and the multiple sampling points across field facilities, a dedicated system was developed to collect, safeguard, and disseminate these results, with a strong emphasis on operational efficiency. To facilitate this, a specialized website and database system were implemented alongside a dedicated workflow to support the team's sampling requests and result-capturing efforts.

To further enhance workflow efficiency, a streamlined automated process was adopted. Sampling requests were initiated via the web platform and subsequently routed to vendors through a SharePoint list. Vendors coordinated with the company's team to schedule site visits. Upon collection and analysis of samples, results were reported back through the same electronic framework, enabling automatic updates to the H₂S database. Throughout the workflow, to maintain the integrity and security of the data, access to request samples and input results is restricted to authorized users only. Additionally, the H₂S database is routinely backed up to ensure long-term data storage and continuity. This integrated process ensures comprehensive coverage of all target wells and provides the capability to reliably track missed site visits or unsuccessful collections for follow-up and resolution.

The system also offers advanced reporting features, allowing users to identify wells lacking recent H₂S data, track performance metrics over time, highlight high-risk wells, and analyze overall trends in H₂S levels across the field. Maintaining a reliable and up-to-date database not only fulfills the specific needs of this project but also aligns with the company's broader Health, Safety, and Environmental (HSE) objectives.

H₂S ROE Calculation Methodology. The ROE is a critical parameter in assessing the potential impact of H₂S dispersion resulting from well blow-outs, and flowlines' and processing lines' loss of containment. The ROE calculation is essential for determining safety zones, defining access protocols, conducting work safely, and planning and implementing appropriate emergency response measures to protect personnel and the environment.

The ROE calculation was completed for three H₂S levels as per the following Levels of Concerns and associated significance:

- 700 ppm: "Knockdown" or immediately collapse.
- 100 ppm: Immediately Dangerous for Life and Health (IDLH). The maximum airborne concentration below which it is believed that nearly all individuals could be exposed for up to 1 hour without experiencing or developing life-threatening effects.
- 10 ppm: 8-hours TWA.

Associated with the above levels of concern, different H₂S zones were defined based on ROE contours for 700 ppm (amber zone), 100 ppm (yellow zone), and 10 ppm (gray zone).

Commercially available Hazard Analysis Software Tool (PHAST) was used to calculate the H₂S ROE. The key factors considered in the ROE calculation include:

- **Source Strength:** The rate of H₂S release from the well blow-out or processing lines. For the well blow-out scenario, the gas rate was calculated by a reservoir engineer and the worst-case potential H₂S was used in the ROE calculation. For processing line with high water content, to determine the gas flow rate from a multiphase leak scenario, the following method was applied:

- Conduct a dispersion study for a 2-inch hole leak originating from the pressure vessel module.
- Record the total mass flow rate for the 2-inch hole.
- Multiply this flow rate by the flashed gas fraction, as determined through HYSYS modeling. This calculated value represents the gas flow rate from the 2-inch hole.
- Use this gas flow rate in the "User Defined Source" scenario to establish the H₂S dispersion contour.
- **Meteorological Conditions:** Conservative Pasquill's atmospheric stability class and wind speed used are 1.5F (stable conditions) and 5D.
- **Release direction:** Horizontal and toward the receptor.

For the well blowout scenario, ROE was calculated for each reservoir based on the potential worst-case H₂S concentration. For flowlines and processing line loss of containments, a table was developed to represent various possible H₂S readings from leak sources. The result of ROE calculation was applied in the development of the H₂S Field Navigator zone classification.

Putting it all together: the "H₂S Field navigator" Application. The "H₂S Field Navigator" smartphone application enables users to select a specific well and provides the H₂S concentration range based the highest historical H₂S readings for that well, specifying the type of measurement (casing vs. flowline). In the absence of H₂S readings, the application issues a cautionary warning to operators. Additionally, the application visually maps risk areas or H₂S zones based on ROE contours for 700 ppm (amber zone), 100 ppm (yellow zone), and 10 ppm (gray zone).

Awareness of H₂S exposure is critical to ensuring personnel safety. A core feature of the application is the integration of H₂S exposure data to dynamically visualize concentrations and their associated risk zones using a combination of automated data ingestion, exposure classification, and empirical dispersion modeling.

An automated process retrieves daily the latest H₂S readings from the measurement database. For each well, the system captures the highest recorded value per component—wellhead, flowline, casing, and HPT gas outlet—and updates a centralized list. As a result, only one record per well component is maintained.

Upon selecting a well in the application, the latest H₂S readings are fetched and categorized into three exposure ranges: less than 10 ppm, between 10 and 100 ppm, and greater than 700 ppm. These ranges are displayed in a color-coded container: red for any detected concentration above zero, green for zero, and black with a precautionary message if no data is available. This immediate visual feedback allows users to assess risk instantly.

Furthermore, to enhance hazard awareness, a ROE model has been integrated into the application. A reference list maps concentration values (e.g., 5,000 ppm to 50,000 ppm) to estimated exposure distances for 10 ppm, 100 ppm, and 700 ppm thresholds. When a user activates the ROE view, the system identifies the highest available reading—prioritizing flowline and wellhead data—and selects the corresponding ROE mapping. The logic ensures that if a reading falls between two thresholds, the higher threshold is used to determine the ROE.

The ROE visualization overlays three concentric zones on the map: gray for 10 ppm, yellow for 100 ppm, and amber for 700 ppm. These zones are dynamically scaled by converting meters to pixels based on the map's current zoom level. A zoom feature recalculates the meters-per-pixel ratio in real time, ensuring accurate and consistent representation of exposure zones at all scales.

The efforts did not stop at well-level monitoring. The application was further enhanced to include H₂S risk classification for processing facilities. For these facilities, a separate reference list is maintained, mapping each facility to a predefined classification: high sour facility, medium sour facility, or low sour facility. When

a user selects a facility, the application retrieves its classification and dynamically renders a corresponding risk zone on the map. The visual representation follows a standardized color scheme: amber for high sour facilities, yellow for medium, and gray for low. This feature extends the scope of H₂S awareness beyond wellheads, enabling users to visualize and assess potential exposure risks around critical infrastructure with the same level of precision and immediacy.

Figure 1 illustrates the data flow, processing logic, and visualization of the "H₂S Field Navigator" application. Coordinates, H₂S measurements, ROE modeling results, and facility risk classifications are ingested from multiple databases. The system processes this data to classify H₂S levels, calculate exposure radii, and define facility risk zones. Map rendering and color-coded risk indicators are generated automatically, with ROE zones displayed as concentric circles.

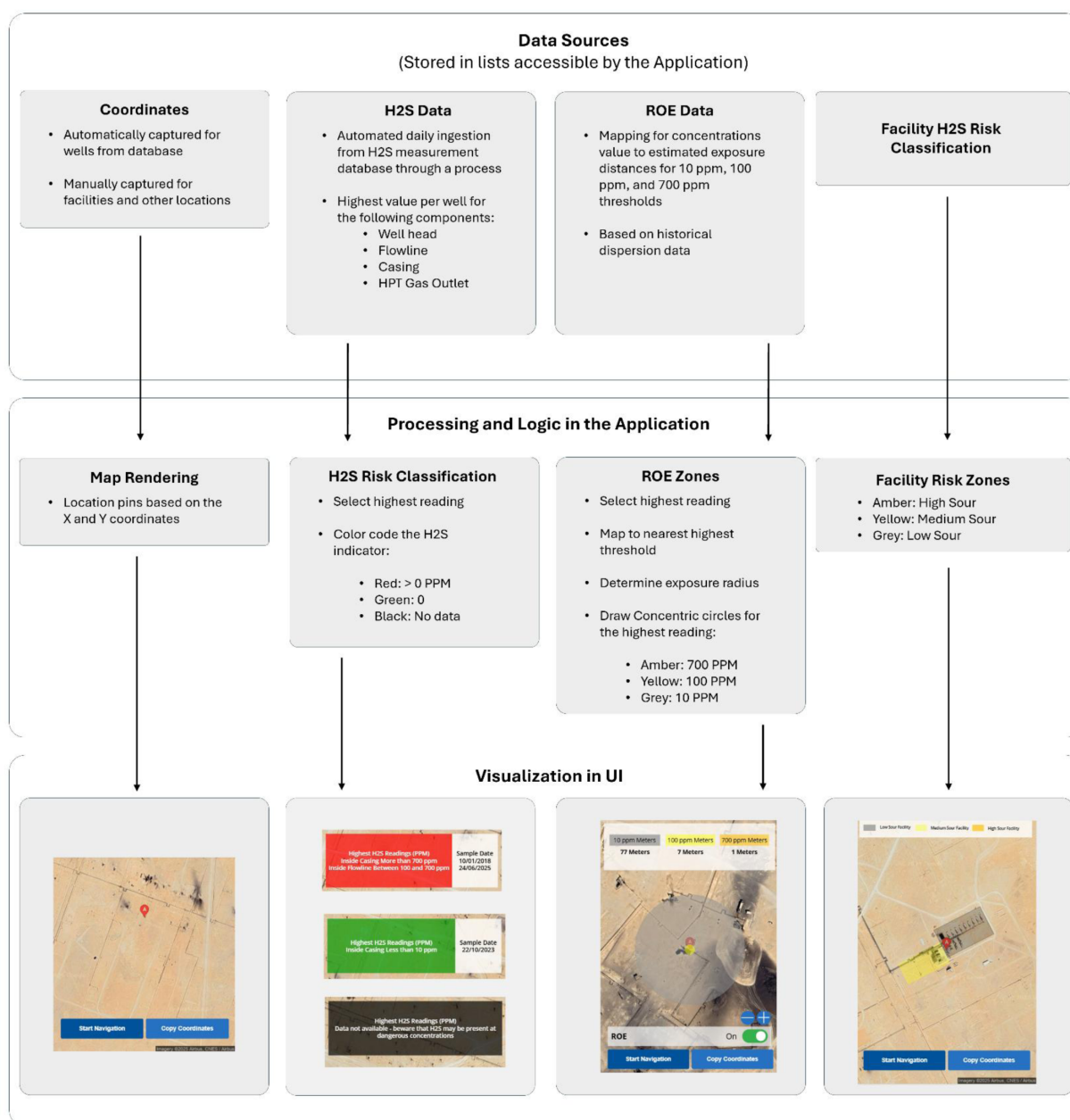


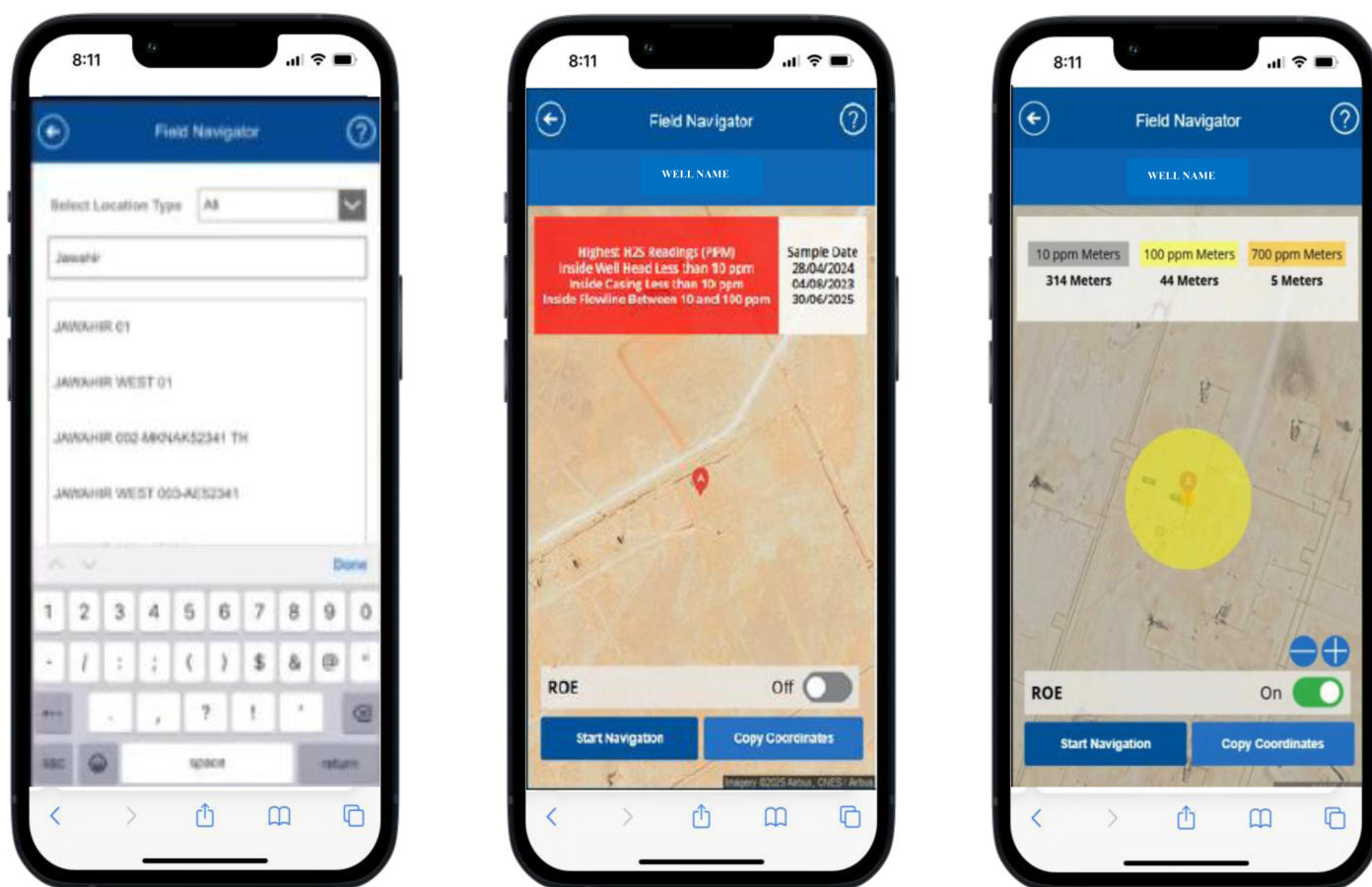
Figure 1—Overview of the Data Sources

Figure 2 presents representative screenshots of the H₂S Field Navigator application, illustrating its core functionalities designed for field operations. The user interface integrates interactive mapping, risk classification, and robust navigation features, thereby facilitating rapid access to location-specific H₂S safety data via mobile devices. Figure 2(a) depicts the well selection interface, enabling users to efficiently identify sites of interest. Figure 2(b) demonstrates the presentation of H₂S measurement results for individual wells, ensuring that operators are informed of critical safety metrics. Figure 2(c) displays the mapping of ROE, allowing for clear visualization of potential risk zones directly within the application.

2(a) User's interface for well selection

2(b) Well's H₂S measurements results

2(c) Radius of Exposure

Figure 2—Different screenshots of the H₂S Field Navigator App

This integrated approach provides a robust framework for H₂S risk visualization. By combining automated data flows, empirical modeling, and dynamic geospatial rendering, the solution empowers field personnel to make informed decisions and maintain safe distances from potential exposure sources. It represents a significant advancement in operational safety and situational awareness in hazardous environments.

Development, Deployment Strategy, and User Adoption

The H₂S Field Navigator encompasses several integral components designed to enhance operational efficiency and field safety. The Navigation Tool was initially developed during the second half of 2023 and reached full deployment in January 2024, where it achieved immediate and widespread acceptance among field operators.

Subsequent integration efforts with the H₂S Measurements Database and the associated ROE were executed in the latter half of 2024. A test version of this integrated platform was released in December 2024, with access restricted to supervisory personnel, HSE professionals, and a select group of field operators. The feedback obtained from these trial users was overwhelmingly positive and instrumental in guiding targeted refinements and calibration.

Following this iterative improvement process, the fully operational H₂S Field Navigator was made accessible to all personnel in the first week of August. To ensure effective implementation and user competency, a dedicated training video was developed, facilitating onboarding and promoting best practices in application usage.

Implementation and Operational Benefits

The development of the "H₂S Field Navigator" smartphone application represents a significant advancement in the oil and gas industry, particularly in the management of H₂S risks.

This project has profound business implications, enhancing operational safety, efficiency, and overall productivity. One of the primary business impacts of the H₂S Field Navigator project is the enhancement of operational safety.

The Field Navigation component offers numerous benefits, including significant time savings previously spent searching for locations. It enables teams to increase the number of completed work orders and activities in the field daily. The application reduces driving distances and the number of road maneuvers, such as turns and returns. Consequently, it lowers vehicle fuel consumption, minimizes wear and tear and reduces the risk exposure of the drivers. Operational costs are decreased through more efficient use of company resources. Road safety is improved by minimizing unnecessary driving and confusion. The application is suitable for all company field teams, including operations, maintenance, contractors, HSE, and emergency response.

The time required to find locations can be reduced from over 30 minutes to approximately 15 minutes for typical trips. The distance driven is anticipated to drop by more than 50% on average, with fewer road turns. Teams can complete more work orders or activities per day, thereby increasing overall productivity. A decrease in traffic incidents and vehicle accidents is expected due to more systematic navigation.

Traditionally, risk areas or H₂S zones were confined to risk study reports or field posters, which were often outdated and not readily accessible. By revolutionizing this approach, the application ensures that field operators have immediate access to critical safety information directly on their smartphones.

According to Oxy Oman's policies, each H₂S zone necessitates distinct safety controls. This is particularly important for the field, which is characterized by wells and reservoirs with a wide variety of risk profiles. For example, the access to areas potentially affected by IDLH concentrations (100 ppm) requires carrying and escape set. The utilization of this smartphone application enhances the implementation of risk-based H₂S policies, improves field operators' access to critical safety information, and ultimately fosters an H₂S culture grounded in awareness and preparedness.

Rapid identification of support facilities significantly enhances emergency response efficiency and expedites search and rescue efforts. Additionally, providing ROE maps via smartphone technology enables real-time visualization of hazardous zones during a potential loss of containment event, thereby facilitating informed decision-making and improving overall safety management.

This real-time data accessibility is crucial for maintaining a safe working environment, as it allows operators to make informed decisions and take necessary precautions to protect themselves and their colleagues.

The application consolidates all H₂S data from periodic field measurement surveys into a centralized database, supported by interactive data visualization software. This centralized approach eliminates the inefficiencies associated with manually searching for and interpreting H₂S data from various sources.

Conclusions

The use of risk areas or H₂S zones is a well-established practice in the oil and gas industry. However, the resulting maps are typically confined to risk study reports or, at best, to field posters. Furthermore, these maps are static and not kept updated with the most recent field measures. The "H₂S Field Navigator" application revolutionizes this by providing the most up-to-date and relevant information directly on smartphones, ensuring immediate accessibility for field operators.

The "H₂S Field Navigator" application significantly enhances the implementation of risk-based H₂S policies by providing real-time data on H₂S concentrations and risk zones. This ensures that field operators have immediate access to critical safety information, which is essential for maintaining a safe working environment. The application consolidates H₂S data from periodic field measurement surveys into a centralized database, supported by interactive data visualization software. This allows for the seamless integration of wellhead locations and process facilities on a web mapping platform based on their coordinates.

The ROE is determined through empirical dispersion modeling using the highest historical readings, and the application updates this data every four hours to ensure the most recent measurements are considered. This dynamic approach ensures that the information provided to field operators is always current and reliable.

In conclusion, the "H₂S Field Navigator" application represents a significant advancement in operational safety and situational awareness in hazardous environments. By providing accurate, up-to-date information directly on smartphones, it ensures immediate accessibility for field operators, enhances the implementation of risk-based H₂S policies, and fosters a culture of awareness and preparedness.

Acknowledgments

The development of this project is the result of the active collaboration of the Company's Field Operations, Reservoir Management, Health, Safety and Environment (HSE), Risk, and Information Technology (IT) departments.